DRAG-HOOKS—Must be attached to each side of gate for opening and closing chains. The hooks to be forged in connection with a plate and secured to the gate and each other by two bolts $\frac{7}{8}$ of an inch diameter. The form and dimensions are fully shown on drawing.

VALVES AND VALVE GEAR.—Three valves to be provided for each gate, each valve having an orifice of discharge $2\frac{1}{2}$ x 5 feet when full open. The three openings to be formed by the insertion of four cast iron chambers containing the controlling motion of the valves. The two centre castings will each contain the motion for the ends of two valves and have an opening through the middle to allow the tie-rods of the gate to pass through them. These castings are intended to serve as frames for the valves, and the two middle ones as mullions or distance-pieces between the bars. The greatest care must, therefore, be exercised in fitting them to the timber in order that the valves may be truly parallel and tight when closed, as well as to ensure a perfectly parallel motion when working.

SHUTTERS.—The two cast iron shutters, which, together, form the valve, have each a wrought iron rod $2\frac{1}{2}$ inches diameter running through them. These rods have their ends turned down to receive brass ferrules 6 inches long and $\frac{1}{8}$ of an inch thick; ferrules to be turned up and tightly fitted on. It will be seen that the rods are extended sufficiently beyond the ends of the shutters to connect them to the motion-links, into which they will be properly fitted. The pivoting side of the shutters will rest in semi-circular sinkings formed on cast iron plates let into and secured to the upper and lower bars of the gate by wood-screws $\frac{3}{4}$ of an inch diameter and 5 inches long. The sinking must be perfectly level and smooth finished. Shutters must have the ends planed or turned up perfectly square with the axis of the pivot, be bored out for the wrought iron rods and brass pivot-pins, have ribs or slots cast in or worked on them, and, when complete, the lines and curves must accurately correspond with those shown on the drawings. The meeting faces of the shutters may have a dovetailed sinking cast in them, with white oak liner and leather packing, or if directed, they may be cast so as to have a smooth finished meeting face $2\frac{1}{2}$ inches wide; if the latter is directed, iron must be provided on casting for finishing up.

PARALLEL MOTION AND CAST IRON CHAMBERS.—The form and dimensions for the cast iron chambers is fully shown on the enlarged drawing. Iron must be left on the casting, so that the finished faces marked "*a*, *a*, *a*, *a*" on drawing, may stand up to the dimensions given when finished. These faces and the pin holes must be at right angles to each other, and templates must be provided for setting out the work, so that all the castings may be true duplicates and interchangeable.

PIVOT PINS—For the arms to be 3 inches diameter and tightly fitted into casting, and finish flush with fitting-strip. The pins for the shutters are to be brass, $2\frac{1}{2}$ inches diameter and 17 inches long. The balancing arms will each be made up of two pieces, sunk and riveted to a cast iron hub. The ends will be bushed with brass $\frac{3}{8}$ of an inch thick, and the hub bushed with brass $\frac{1}{2}$ an inch thick; a fitting ring of brass must be secured on to face of hub by screws $\frac{1}{4}$ of an inch diameter, tapped into the casting. Wrought iron links 20 inches long and 16 inches centres will be attached to arms by pins 2 inches diameter, pin to be tightly fitted to link and work in the brasses of the arm. These links will all require to be exact duplicates of each other and interchangeable. The only finish required, if they are properly forged in dies, will be the boring out for pins.

If a sufficiently tight joint cannot be made between hub and diaphram *b*, the casting must be bored out, and a brass pin $\frac{3}{4}$ of an inch diameter, inserted as shown at *c*, on drawing.